AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 4: State-of-health estimation models

Mon 23 Nov - Fri 27 Nov 2026 · Sprint 4 · Overall progress 42%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 27 Nov 2026

On track **This week in one line:** Five models compared on cells never seen in training. SOH error 2.20% (NASA, Gradient Boosting) and 0.81% (CALCE, Gaussian Process), with 95% intervals covering 95-98% of true values.

1. Goal for the week

Estimate SOH from charging signals only, validated honestly on unseen cells, with calibrated uncertainty.

2. Day-by-day work

Day	Work done
Mon 23 Nov	Leave-one-cell-out harness: each cell is scored by a model trained only on the other three cells of its dataset.
Tue 24 Nov	Trained Ridge, Random Forest, Gradient Boosting, MLP and Gaussian Process; compared relative vs absolute features.
Wed 25 Nov	Switched to absolute features (NASA error fell from 4.1% to 2.2%); added the partial-charge feature set.
Thu 26 Nov	GP intervals were over-confident (66% coverage); replaced with split-conformal 95% intervals.
Fri 27 Nov	Saved production models, figure, 4 new tests (16 total), this report.

3. Completed deliverables

- **batterylife/models/soh.py** and evaluation.py.
- **results/soh_metrics.json**, soh_predictions.csv.
- tests/test_soh.py.

4. Results

Model	NASA full	NASA partial	CALCE full	CALCE partial
Ridge	2.35	2.96	0.83	5.87
Random Forest	2.41	2.49	1.15	1.96
Gradient Boosting	2.20	2.49	1.32	2.07
MLP	5.01	7.07	7.40	3.28
Gaussian Process	3.82	4.20	0.81	3.83

RMSE in SOH percentage points on unseen cells. 'Partial' leaves out CC charge time (charge may start part-way).

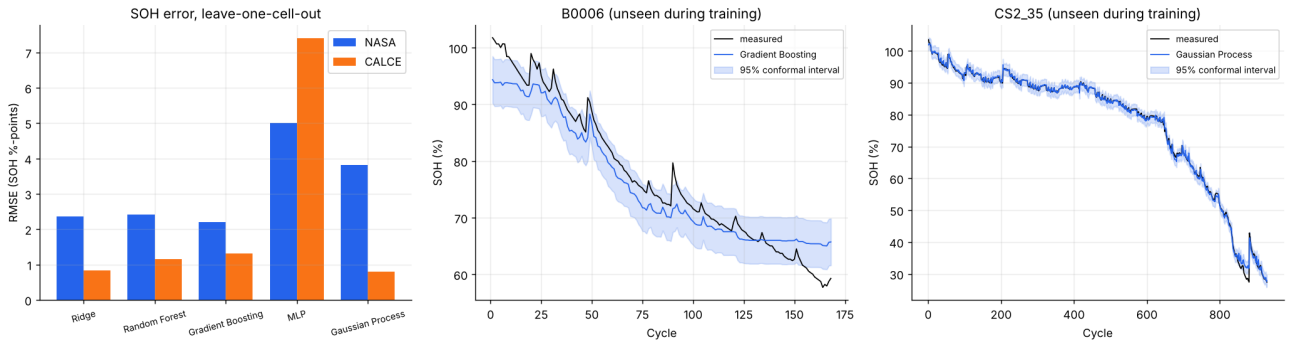


Figure 4. Left: error by model. Middle and right: estimate and 95% interval on two unseen cells.

5. Problems faced and how they were solved

Problem	Resolution
Relative features erased differences in starting capacity between NASA cells.	Absolute features adopted.
Tree models cannot extrapolate below the training range (B0006 at 58%).	Documented; more deep-fade cells would fix it.
MLP underperforms with only 3 training cells.	Kept only as a baseline.

6. Plan for next week

- Predict remaining useful life to 80% and 60% SOH.
- Calibrate the uncertainty bands.

Faculty remarks

Faculty signature: _____ Student signature: _____